Chapter 2

Curves in State Space

The shortest distance between two points is a straight line, but no interesting motion has ever been straight.

2.1 Why Geometry Comes First

In Chapter 1 we argued that control is fundamentally about *motion*—the faithful execution of a trajectory through state space. Before we can stabilize a trajectory, optimize a trajectory, or confine the system to a tube around a trajectory, we need to understand what a trajectory *is* as a mathematical object.

This chapter develops the differential geometry of curves in $\mathbb{R}^n$, adapted to the needs of control theory. The central insight is that a trajectory has intrinsic geometric properties—curvature, torsion, and a natural moving frame—that exist independently of how fast the system moves along the curve. These intrinsic properties determine how difficult the trajectory is to track, how much control authority is needed to stay on course, and where the system is most vulnerable to disturbances.

The reader may wonder: *why not simply work in coordinates?* The answer is that coordinate-based descriptions mix up properties of the *curve* with properties of the *parameterization.* A golf swing executed slowly and the same swing executed quickly trace the same curve in configuration space, but their coordinate descriptions $\boldsymbol{q}(t)$ are entirely different functions. The geometric tools we develop here allow us to separate what is intrinsic to the motion from what is merely a choice of clock.

Key Idea 2.1. Intrinsic vs. Extrinsic

A trajectory has two kinds of properties:

- **Intrinsic**: properties of the curve itself—its shape, curvature, how it bends and twists through state space. These are independent of parameterization.
- **Extrinsic**: properties that depend on how the curve is traversed—speed, acceleration, timing. These depend on the parameterization.

Good control design separates these cleanly. The *trajectory planner* designs the intrinsic shape. The *timing law* specifies the extrinsic speed profile. The *tracking controller* maintains both.

2.2 Parameterized Curves

Definition 2.1. Parameterized Curve

A **parameterized curve** in $\mathbb{R}^n$ is a smooth map

$$\boldsymbol{\gamma} : I \to \mathbb{R}^n, \qquad \lambda \mapsto \boldsymbol{\gamma}(\lambda), \tag{2.1}$$

where $I \subseteq \mathbb{R}$ is an interval. The parameter λ may be time t, arc length s, a phase variable σ, or any other monotone scalar. The curve is **regular** if $\boldsymbol{\gamma}'(\lambda) \neq \mathbf{0}$ for all $\lambda \in I$, i.e., it never "stops" (though the system traversing it may slow down).

The critical point is that many different parameterizations describe the same geometric curve. If $\boldsymbol{\gamma}(\lambda)$ is a parameterized curve and $\lambda = \phi(\mu)$ is a smooth, monotonically increasing reparameterization, then $\tilde{\boldsymbol{\gamma}}(\mu) = \boldsymbol{\gamma}(\phi(\mu))$ traces the same set of points in $\mathbb{R}^n$, but at different "speeds."

Example 2.1. The Circle: Three Parameterizations

Consider the unit circle in $\mathbb{R}^2$. Three natural parameterizations:

$$\boldsymbol{\gamma}_1(t) = (\cos t,\ \sin t), \quad t \in [0, 2\pi] \qquad \text{(constant angular speed)}, \tag{2.2}$$

$$\boldsymbol{\gamma}_2(t) = (\cos t^2,\ \sin t^2), \quad t \in [0, \sqrt{2\pi}] \qquad \text{(accelerating)}, \tag{2.3}$$

$$\boldsymbol{\gamma}_3(s) = (\cos s,\ \sin s), \quad s \in [0, 2\pi] \qquad \text{(arc-length, same as } \boldsymbol{\gamma}_1 \text{ here)}. \tag{2.4}$$

All three trace the same circle. The *curve* is the same; the *parameterizations* differ. Quantities like curvature are properties of the curve, not the parameterization.

2.2.1 Velocity, Speed, and the Tangent Vector

Given a parameterized curve $\boldsymbol{\gamma}(\lambda)$, the **velocity vector** is

$$\boldsymbol{v}(\lambda) = \frac{\mathrm{d}\boldsymbol{\gamma}}{\mathrm{d}\lambda} = \boldsymbol{\gamma}'(\lambda). \tag{2.5}$$

The **speed** is its magnitude:

$$v(\lambda) = \left\|\boldsymbol{\gamma}'(\lambda)\right\|. \tag{2.6}$$

The **unit tangent vector** is the velocity normalized to unit length:

$$\boldsymbol{e}_1(\lambda) = \frac{\boldsymbol{\gamma}'(\lambda)}{\|\boldsymbol{\gamma}'(\lambda)\|} = \frac{\boldsymbol{\gamma}'(\lambda)}{v(\lambda)}. \tag{2.7}$$

The tangent vector $\boldsymbol{e}_1$ tells us the *direction* of the curve at each point—a purely geometric quantity. The speed v tells us how *fast* we traverse it—a parameterization-dependent quantity. This separation is the first instance of the intrinsic/extrinsic decomposition.

2.3 Arc Length: The Natural Parameter

Among all possible parameterizations, one is geometrically privileged: the **arc-length parameterization**, in which the parameter measures distance traveled along the curve.

Definition 2.2. Arc Length

The **arc length** of a curve $\boldsymbol{\gamma}(\lambda)$ from $\lambda = a$ to $\lambda = b$ is

$$L = \int_a^b \|\boldsymbol{\gamma}'(\lambda)\| \, \mathrm{d}\lambda = \int_a^b v(\lambda) \, \mathrm{d}\lambda. \tag{2.8}$$

The **arc-length function** $s(\lambda)$ measures the distance from the starting point:

$$s(\lambda) = \int_a^\lambda \|\boldsymbol{\gamma}'(\mu)\| \, \mathrm{d}\mu, \qquad \text{so that } \frac{\mathrm{d}s}{\mathrm{d}\lambda} = v(\lambda). \tag{2.9}$$

When a curve is parameterized by arc length, the speed is identically 1: $\|\boldsymbol{\gamma}'(s)\| = 1$ for all s. This means the tangent vector and the velocity vector coincide: $\boldsymbol{e}_1(s) = \boldsymbol{\gamma}'(s)$. All geometric information is in the direction of the derivative; none is hidden in its magnitude.

Principle 2.1. Arc Length as the Canonical Clock

Arc-length parameterization strips away all timing information and reveals the pure geometry of the curve. In control terms, it answers the question: *what does the motion look like if we forget how fast it goes?*

For a golf swing, the arc-length parameterized trajectory $\boldsymbol{\gamma}(s)$ describes the spatial *path* of the body through configuration space. The timing law $s(t)$—how the golfer moves along this path in real time—is a separate design choice. A slow practice swing and a full-speed competition swing follow the same $\boldsymbol{\gamma}(s)$ with different $s(t)$.

Remark 2.1. Arc Length in High Dimensions

In $\mathbb{R}^n$ with the Euclidean metric, arc length is defined exactly as above. When the state space carries a non-Euclidean metric (as it will when we work on configuration manifolds in Chapter 3), the formula (2.8) generalizes to

$$L = \int_a^b \sqrt{\boldsymbol{\gamma}'(\lambda)^\top \boldsymbol{G}(\boldsymbol{\gamma}(\lambda))\, \boldsymbol{\gamma}'(\lambda)} \, \mathrm{d}\lambda, \tag{2.10}$$

where $\boldsymbol{G}(\boldsymbol{x})$ is the Riemannian metric tensor (for mechanical systems, this is the mass-inertia matrix). This means "distance in configuration space" naturally accounts for how massive different parts of the body are. A degree of freedom connected to a heavy segment contributes more to arc length than one connected to a light segment—exactly as physical intuition suggests.

2.4 Curvature: How Curves Bend

The most important intrinsic property of a curve is its **curvature**—a measure of how rapidly the tangent direction changes as we move along the curve.

Definition 2.3. Curvature

Let $\boldsymbol{\gamma}(s)$ be a curve parameterized by arc length in $\mathbb{R}^n$. The **curvature** at each point is

$$\kappa(s) = \|\boldsymbol{\gamma}''(s)\| = \left\| \frac{\mathrm{d}\boldsymbol{e}_1}{\mathrm{d}s} \right\|. \tag{2.11}$$

Since $\|\boldsymbol{e}_1\| = 1$, the derivative $\boldsymbol{e}_1'(s)$ is orthogonal to $\boldsymbol{e}_1(s)$, and κ measures the rate of turning per unit distance. The curvature is always non-negative and is zero if and only if the curve is locally a straight line.

The reciprocal $R(s) = 1/\kappa(s)$ is the **radius of curvature**—the radius of the circle that best approximates the curve at that point. High curvature means a tight bend; low curvature means a gentle sweep.

2.4.1 Curvature in Arbitrary Parameterization

In practice, we rarely work in arc-length parameterization. Given a curve $\boldsymbol{\gamma}(t)$ parameterized by time (or any other parameter), the curvature can be computed directly:

Theorem 2.1 (Curvature Formula)**.** *For a regular curve $\boldsymbol{\gamma}(t)$ in $\mathbb{R}^n$, the curvature is*

$$\kappa(t) = \frac{\|\boldsymbol{\gamma}'(t) \times_n \boldsymbol{\gamma}''(t)\|}{\|\boldsymbol{\gamma}'(t)\|^3}, \tag{2.12}$$

where for $n = 2$, $\|\boldsymbol{\gamma}' \times_2 \boldsymbol{\gamma}''\| = |x'y'' - y'x''|$, and for $n = 3$, $\times_3$ is the usual cross product. For general n, this generalizes via

$$\kappa(t) = \frac{\sqrt{\|\boldsymbol{\gamma}'\|^2 \|\boldsymbol{\gamma}''\|^2 - (\boldsymbol{\gamma}' \cdot \boldsymbol{\gamma}'')^2}}{\|\boldsymbol{\gamma}'\|^3}. \tag{2.13}$$

Proof. Write $\boldsymbol{\gamma}(t) = \boldsymbol{\gamma}(s(t))$ where s is arc length. Then $\boldsymbol{\gamma}' = \dot{s}\,\boldsymbol{e}_1$ and $\boldsymbol{\gamma}'' = \ddot{s}\,\boldsymbol{e}_1 + \dot{s}^2\,\kappa\,\boldsymbol{e}_2$, where $\boldsymbol{e}_2$ is the unit normal (defined below). Since $\boldsymbol{e}_1 \perp \boldsymbol{e}_2$, we have $\|\boldsymbol{\gamma}'\|^2 \|\boldsymbol{\gamma}''\|^2 - (\boldsymbol{\gamma}' \cdot \boldsymbol{\gamma}'')^2 = \dot{s}^2(\ddot{s}^2 + \dot{s}^4\kappa^2) - \dot{s}^2\ddot{s}^2 = \dot{s}^6\kappa^2$. Dividing by $\|\boldsymbol{\gamma}'\|^6 = \dot{s}^6$ and taking the square root yields the result. $\square$

Example 2.2. Curvature of an Elliptical Trajectory

Consider the ellipse $\boldsymbol{\gamma}(t) = (a\cos t,\ b\sin t)$ with $a > b > 0$. Then

$$\boldsymbol{\gamma}'(t) = (-a\sin t,\ b\cos t), \tag{2.14}$$
$$\boldsymbol{\gamma}''(t) = (-a\cos t,\ -b\sin t). \tag{2.15}$$

Applying the 2D curvature formula:

$$\kappa(t) = \frac{|(-a\sin t)(-b\sin t) - (b\cos t)(-a\cos t)|}{(a^2\sin^2 t + b^2\cos^2 t)^{3/2}} = \frac{ab}{(a^2\sin^2 t + b^2\cos^2 t)^{3/2}}. \tag{2.16}$$

The curvature is *highest at the ends of the minor axis* ($t = 0, \pi$, where $\kappa = a/b^2$) and *lowest at the ends of the major axis* ($t = \pi/2, 3\pi/2$, where $\kappa = b/a^2$). The tighter the bend, the higher the curvature—and, as we will see, the more control authority is needed to track it.

2.4.2 Why Curvature Matters for Control

Curvature is not merely a geometric curiosity. It has direct implications for tracking difficulty.

Principle 2.2. The Curvature–Authority Principle

Consider a system tracking a reference trajectory $\boldsymbol{\gamma}^*(s)$ at speed v. The centripetal acceleration required to follow the curve is

$$a_\perp = \kappa v^2. \tag{2.17}$$

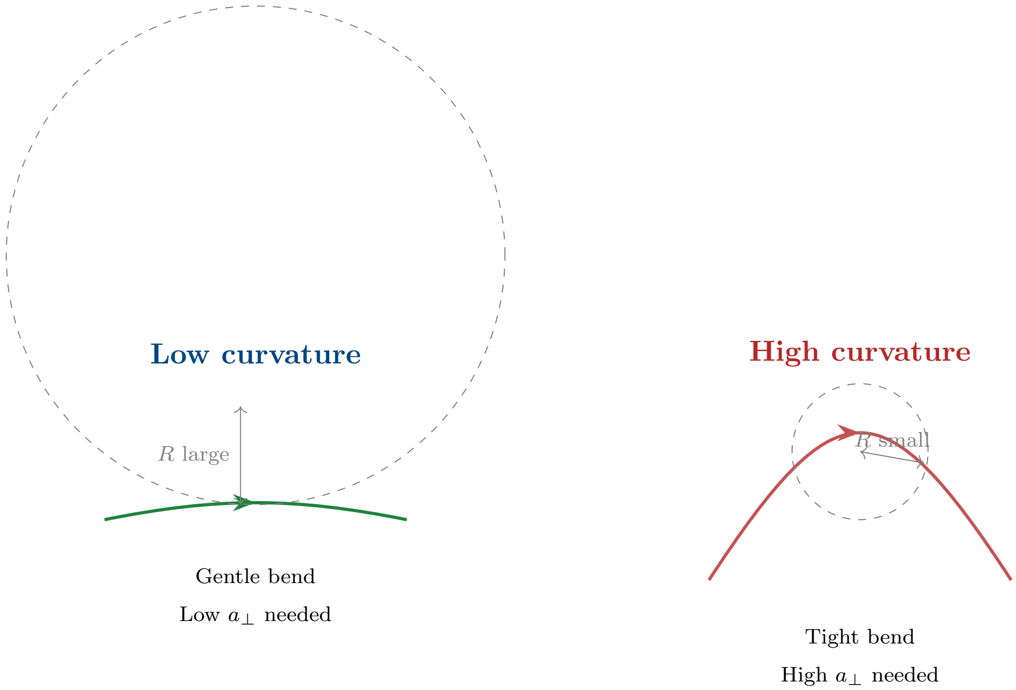
This means:

(i) High curvature demands high lateral control authority.

(ii) The required authority grows as the *square* of the speed.

(iii) A trajectory that is easy to track slowly may be impossible to track at high speed.

For the golf swing, this explains why the transition from backswing to downswing is so critical: the clubhead path has high curvature precisely where the speed is increasing rapidly.



2.5 The Frenet–Serret Frame

The tangent vector $\boldsymbol{e}_1$ is only the first element of a natural coordinate frame that moves with the curve. This **Frenet–Serret frame** (or simply **moving frame**) provides a complete local coordinate system at every point of the trajectory.

2.5.1 Construction in $\mathbb{R}^n$

Definition 2.4. Frenet–Serret Frame

Let $\boldsymbol{\gamma}(s)$ be a curve parameterized by arc length in $\mathbb{R}^n$ with $\boldsymbol{\gamma}^{(k)}(s)$ denoting the k-th derivative with respect to s. Assume the first $p \leq n$ derivatives $\{\boldsymbol{\gamma}'(s), \boldsymbol{\gamma}''(s), \ldots, \boldsymbol{\gamma}^{(p)}(s)\}$ are linearly independent at each point. The **Frenet–Serret frame** $\{\boldsymbol{e}_1(s), \boldsymbol{e}_2(s), \ldots, \boldsymbol{e}_p(s)\}$ is defined by applying the Gram–Schmidt process to these derivatives:

$$\boldsymbol{e}_1 = \boldsymbol{\gamma}' = \boldsymbol{e}_1, \tag{2.18}$$

$$\tilde{\boldsymbol{e}}_k = \boldsymbol{\gamma}^{(k)} - \sum_{j=1}^{k-1} \left\langle \boldsymbol{\gamma}^{(k)}, \boldsymbol{e}_j \right\rangle \boldsymbol{e}_j, \qquad \boldsymbol{e}_k = \frac{\tilde{\boldsymbol{e}}_k}{\|\tilde{\boldsymbol{e}}_k\|}, \quad k = 2, \ldots, p. \tag{2.19}$$

The frame vectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_p$ form an orthonormal set at each point. If $p < n$, the frame can be extended to a full orthonormal basis by appending $n - p$ vectors, but these additional vectors have no intrinsic geometric meaning for the curve.

For the familiar case $n = 3$:

Vector	**Name**	**Interpretation**
$\boldsymbol{e}_1 = \boldsymbol{e}_1$	Unit tangent	Direction of motion
$\boldsymbol{e}_2 = \boldsymbol{e}_2$	Principal normal	Direction the curve is bending toward
$\boldsymbol{e}_3 = \boldsymbol{e}_3$	Binormal	$\boldsymbol{e}_1 \times \boldsymbol{e}_2$; normal to the osculating plane

2.5.2 The Frenet–Serret Equations

The power of the moving frame lies in how it evolves along the curve. The derivatives of the frame vectors with respect to arc length are expressed entirely in terms of the frame itself and a set of scalar functions called **generalized curvatures**.

Theorem 2.2 (Frenet–Serret Equations)**.** *The Frenet–Serret frame satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}s}\begin{pmatrix} \boldsymbol{e}_1 \\ \boldsymbol{e}_2 \\ \boldsymbol{e}_3 \\ \vdots \\ \boldsymbol{e}_p \end{pmatrix} = \begin{pmatrix} 0 & \kappa_1 & 0 & \cdots & 0 \\ -\kappa_1 & 0 & \kappa_2 & \cdots & 0 \\ 0 & -\kappa_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \kappa_{p-1} \\ 0 & \cdots & 0 & -\kappa_{p-1} & 0 \end{pmatrix}\begin{pmatrix} \boldsymbol{e}_1 \\ \boldsymbol{e}_2 \\ \boldsymbol{e}_3 \\ \vdots \\ \boldsymbol{e}_p \end{pmatrix}, \tag{2.20}$$

where $\kappa_1(s) = \kappa(s) > 0$ is the curvature, $\kappa_2(s)$ is the ***torsion*** *(often denoted τ), and $\kappa_3, \ldots, \kappa_{p-1}$ are the* ***higher-order curvatures****. The matrix is skew-symmetric, which guarantees that the frame remains orthonormal as it evolves.*

The skew-symmetry of this matrix is not a coincidence—it is the hallmark of a *rotation*. As the frame moves along the curve, it rotates without stretching. The curvatures $\kappa_1, \ldots, \kappa_{p-1}$ are the "angular velocities" of this rotation projected onto the various planes of the frame.

Example 2.3. Frenet–Serret for a 3D Helix

The helix $\boldsymbol{\gamma}(t) = (a\cos t,\ a\sin t,\ bt)$ with speed $v = \sqrt{a^2 + b^2}$ has arc-length parameter $s = vt$. The Frenet–Serret quantities are:

$$\boldsymbol{e}_1 = \frac{1}{v}(-a\sin t,\ a\cos t,\ b), \tag{2.21}$$

$$\boldsymbol{e}_2 = (-\cos t,\ -\sin t,\ 0), \tag{2.22}$$

$$\boldsymbol{e}_3 = \frac{1}{v}(b\sin t,\ -b\cos t,\ a), \tag{2.23}$$

$$\kappa = \frac{a}{a^2 + b^2}, \qquad \tau = \frac{b}{a^2 + b^2}. \tag{2.24}$$

Both curvature and torsion are *constant*, which makes the helix a natural reference trajectory for testing tracking controllers—analogous to the role of step inputs in classical control. The helix is to trajectory control what the step function is to setpoint control.

2.5.3 The Moving Frame in State Space

For control applications, the state space typically has dimension $n = 2N$ where N is the number of generalized coordinates (positions and velocities). A golf swing model with 4 DOF has $n = 8$. The Frenet–Serret frame in $\mathbb{R}^8$ has up to 7 generalized curvatures, κ_1 through κ_7.

Not all of these are physically meaningful for every system. In practice, the effective dimensionality of the curve—the number of linearly independent derivatives—is determined by the system dynamics:

Principle 2.3. Trajectory Derivatives and Input Rank

The dimension of the input image at one state does not bound the rank of a trajectory's successive derivatives by twice the actuator count. Drift, coupling, and the evolving input can generate additional derivative directions. For the one-input, two-coordinate spring system in Chapter 1, the unforced trajectory from $x(0) = (1, 0, 0, 0)^T$ has $\det[Ax, A^2x, A^3x, A^4x] = -1$. At that regular point its first four time derivatives are independent, and a smooth arc-length reparameterization preserves their rank. A Frenet frame can therefore span four directions even though only one input is available. A single regular trajectory remains a one-dimensional curve. Its local osculating spaces, the state space, and the union of reachable trajectories have different dimensions. Determine the derivative ranks and required regularity for the actual curve; do not infer them from actuator count. Conversely, full actuation does not remove the kinematic compatibility between position and velocity or make an arbitrary state-space curve feasible.

2.6 Curvature and Torsion in the Language of Control

We now connect the geometric quantities to the dynamics and control of nonlinear systems.

2.6.1 Curvature as a Demand on the Controller

Consider the system $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ tracking a reference trajectory $\boldsymbol{\gamma}^*(s)$ at instantaneous speed $v(t) = \dot{s}(t)$. Using the chain rule:

$$\dot{\boldsymbol{x}} = v\,\boldsymbol{e}_1, \qquad \ddot{\boldsymbol{x}} = \dot{v}\,\boldsymbol{e}_1 + v^2\,\kappa\,\boldsymbol{e}_2. \tag{2.25}$$

The acceleration decomposes into:

- A **tangential component** $\dot{v}\,\boldsymbol{e}_1$ that changes the speed along the curve.
- A **normal component** $v^2\kappa\,\boldsymbol{e}_2$ that "steers" the system around the bend.

Definition 2.5. Control Demand Decomposition

Along a reference trajectory $\boldsymbol{\gamma}^*(s)$ traversed at speed v, the instantaneous control demand decomposes as

$$\boldsymbol{u}_{\text{demand}} = \underbrace{\boldsymbol{u}_{\parallel}}_{\text{tangential}} + \underbrace{\boldsymbol{u}_{\perp}}_{\text{normal}}, \tag{2.26}$$

where:

$$\boldsymbol{u}_{\parallel} \propto \dot{v} \qquad \text{(accelerate/decelerate along path)}, \tag{2.27}$$

$$\boldsymbol{u}_{\perp} \propto v^2\,\kappa \qquad \text{(centripetal force to follow curvature)}. \tag{2.28}$$

The normal demand $\boldsymbol{u}_{\perp}$ grows quadratically with speed and linearly with curvature. This is the **curvature cost** of the trajectory.

Example 2.4. Curvature Cost in the Golf Downswing

In the transition from backswing to downswing, a professional golfer reverses the direction of the club in roughly 0.15 seconds. Let us estimate the curvature cost.

At the top of the backswing, the clubhead speed is approximately zero. At impact (0.3 s later), it reaches $\sim 50\,\text{m/s}$. The path through configuration space has a tight reversal—a region of high curvature κ.

Using $a_{\perp} = \kappa v^2$, and noting that mid-downswing $v \approx 25\,\text{m/s}$ with the reversal curvature $\kappa \approx 2\,\text{rad/m}$ (estimated from motion capture data), the centripetal acceleration demand is

$$a_{\perp} \approx 2 \times 25^2 = 1250\ \text{m/s}^2 \approx 127\,g. \tag{2.29}$$

This enormous demand is met not by muscular torque alone, but by the *passive coupling* in the kinematic chain—the very underactuation that the setpoint framework treats as a constraint. The geometry of the trajectory creates the forces needed to execute it.

2.6.2 Torsion as Out-of-Plane Complexity

If curvature measures how a trajectory bends *within* its osculating plane, **torsion** measures how the osculating plane itself rotates along the curve. A planar curve has zero torsion everywhere. A helix has constant nonzero torsion. A golf swing—which involves simultaneous rotation in the transverse, sagittal, and frontal planes—has torsion that varies dramatically through the motion.

Definition 2.6. Torsion

For a curve in $\mathbb{R}^n$ with $n \geq 3$, the **torsion** $\tau(s) = \kappa_2(s)$ is the second generalized curvature in the Frenet–Serret equations:

$$\frac{\mathrm{d}\boldsymbol{e}_2}{\mathrm{d}s} = -\kappa\,\boldsymbol{e}_1 + \tau\,\boldsymbol{e}_3. \tag{2.30}$$

Torsion measures the rate at which the osculating plane rotates about the tangent direction, per unit arc length.

In high-dimensional state spaces ($n \gg 3$), the higher curvatures $\kappa_3, \ldots, \kappa_{p-1}$ measure increasingly subtle geometric twisting. For control design, the first two curvatures (κ and τ) capture the dominant geometric effects, with higher curvatures becoming relevant primarily in highly articulated systems.

Remark 2.2. The Curvature Signature

The ordered set of generalized curvatures $(\kappa_1(s), \kappa_2(s), \ldots, \kappa_{p-1}(s))$ as functions of arc length constitutes the **curvature signature** of the trajectory. By the fundamental theorem of curves, this signature (up to rigid motions) *uniquely determines* the trajectory. Two motions with identical curvature signatures are geometrically congruent—they are the same shape, possibly translated and rotated.
This has a striking implication for motor control: if a golfer can reproduce the curvature signature of their swing, the spatial path is automatically reproduced, regardless of timing.

2.7 Reparameterization and Timing Laws

We now formalize the separation between the *shape* of a trajectory and the *timing* of its execution.

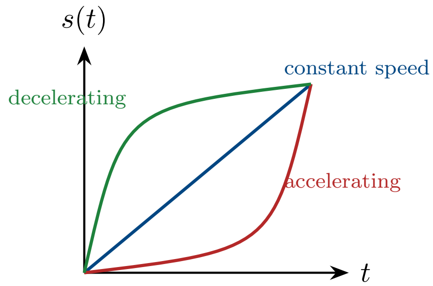
Definition 2.7. Timing Law

Given a reference trajectory $\boldsymbol{\gamma}^*(s)$ parameterized by arc length, a **timing law** is a smooth, monotonically increasing function $s : [0, T] \to [0, L]$ with $s(0) = 0$ and $s(T) = L$. The time-parameterized trajectory is

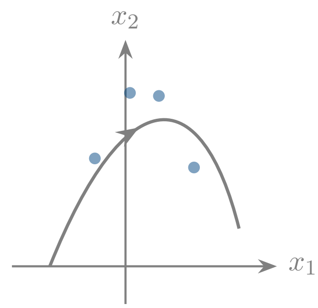
$$\boldsymbol{x}^*(t) = \boldsymbol{\gamma}^*(s(t)). \tag{2.31}$$

The timing law is equivalently specified by the speed profile $v(t) = \dot{s}(t)$ or the phase velocity $v(s) = \mathrm{d}s/\mathrm{d}t$ expressed as a function of arc length.

Different timing laws produce different motions through the same geometric path:



Three timing laws



Same spatial path

2.7.1 The Path–Timing Decomposition of Control

The reparameterization framework leads to a clean decomposition of the control problem:

Principle 2.4. Path–Timing Separation

The trajectory control problem decomposes into two subproblems:

(A) **Path design**: Choose the geometric curve $\boldsymbol{\gamma}^*(s)$ in state space that achieves the task (e.g., delivers the clubhead to the ball with the correct velocity vector).

(B) **Timing design**: Choose the timing law $s(t)$ that balances speed against tracking difficulty, control effort, and robustness.

These two problems have different objective functions and constraints:

	Path design	**Timing design**
Objective	Task performance	Trackability & effort
Key quantity	Curvature $\kappa(s)$	Speed profile $v(s)$
Constraint	Dynamic feasibility	Actuator limits

The interaction between the two is captured by the curvature cost $\kappa(s)\,v(s)^2$, which the timing law must keep within the controller's authority at every point.

Example 2.5. Timing the Downswing

Suppose the downswing path has a curvature profile $\kappa(s)$ with a peak of $\kappa_{\max}$ at the transition point. The maximum centripetal acceleration the controller can provide is $a_{\max}$. Then the speed at the transition is bounded by

$$v_{\text{transition}} \le \sqrt{\frac{a_{\max}}{\kappa_{\max}}}. \tag{2.32}$$

This is a *geometric speed limit*: no matter how powerful the actuators, the system cannot traverse a tight bend faster than its control authority allows. The timing law must respect this limit, typically by slowing down through high-curvature regions and accelerating through straight sections.

This is precisely what golfers do instinctively. The "pause at the top"—the brief deceleration between backswing and downswing—is not aesthetic affectation. It is the timing law respecting the curvature speed limit at the point of maximum path curvature.

2.8 The Osculating Objects

The Frenet–Serret frame defines a sequence of "best-fit" geometric objects at each point of the curve. These osculating objects provide increasingly refined local approximations that are useful for controller design.

Definition 2.8. Osculating Objects

At a point $\boldsymbol{\gamma}^*(s_0)$ on a curve in $\mathbb{R}^n$:

(i) The **osculating line** is the tangent line: the best linear approximation.

(ii) The **osculating circle** lies in the $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ plane, has radius $R = 1/\kappa$, and center at $\boldsymbol{\gamma}^*(s_0) + R\,\boldsymbol{e}_2$. It is the best second-order (circular) approximation.

(iii) The **osculating plane** is the plane spanned by $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. Torsion measures how fast the curve leaves this plane.

(iv) The **osculating sphere** is the best third-order approximation and accounts for both curvature and torsion.

For trajectory control, the osculating circle is the most immediately useful. It tells us: *if the system were constrained to follow a circle at this point, what circle would it be?* The radius of that circle is the radius of curvature $R = 1/\kappa$, and the center indicates the direction toward which the trajectory bends.

2.9 The Geometry of Trajectory Deviations

We now use the Frenet–Serret frame to analyze what happens when the system deviates from the reference trajectory. This provides the geometric foundation for the transverse linearization of Chapter 4.

2.9.1 Frenet–Serret Coordinates

Given a reference trajectory $\boldsymbol{\gamma}^*(s)$ and the associated Frenet–Serret frame $\{\boldsymbol{e}_1(s), \ldots, \boldsymbol{e}_p(s)\}$, any state $\boldsymbol{x}$ near the trajectory can be decomposed as:

$$\boldsymbol{x} = \boldsymbol{\gamma}^*(s^*) + \sum_{k=2}^{n} \xi_k \, \boldsymbol{e}_k(s^*), \tag{2.33}$$

where $s^* = s^*(\boldsymbol{x})$ is the projection phase (the arc-length parameter of the nearest point on the reference), and $\xi_2, \ldots, \xi_n$ are the **transverse coordinates** measuring deviation in each normal direction.

Definition 2.9. Frenet–Serret Coordinates

The **Frenet–Serret coordinate representation** of a state $\boldsymbol{x}$ relative to a reference trajectory $\boldsymbol{\gamma}^*$ is the tuple

$$(s^*, \ \xi_2, \ \xi_3, \ \ldots, \ \xi_n), \tag{2.34}$$

where s^* is the tangential (along-trajectory) coordinate and $\boldsymbol{\xi} = (\xi_2, \ldots, \xi_n)$ are the transverse (cross-trajectory) coordinates. The transverse distance from Chapter 1 is

$$d_\perp = \|\boldsymbol{\xi}\| = \sqrt{\xi_2^2 + \cdots + \xi_n^2}. \tag{2.35}$$

These coordinates have a powerful property: the reference trajectory itself corresponds to $\boldsymbol{\xi} = \mathbf{0}$, and the trajectory-tracking problem becomes the problem of *regulating* $\boldsymbol{\xi}$ *to zero while* s^* *advances*. This converts the tracking problem into a regulation problem, but in the moving frame rather than a fixed frame.

2.9.2 Curvature-Induced Coupling

There is a subtlety: the Frenet–Serret coordinates are not inertial. Because the frame rotates as it moves along the curve (governed by the Frenet–Serret equations (2.20)), fictitious forces appear in the transverse dynamics. The dominant effect is a **curvature-induced coupling**:

Proposition 2.1. *In Frenet–Serret coordinates, the linearized transverse dynamics about the reference trajectory have the form*

$$\dot{\boldsymbol{\xi}} = \big[\boldsymbol{A}(s) - v^2 \boldsymbol{K}(s)\big] \, \boldsymbol{\xi} + \boldsymbol{B}(s) \, \boldsymbol{u}_{fb}, \tag{2.36}$$

where $\boldsymbol{A}(s)$ *comes from linearizing the original dynamics, and* $\boldsymbol{K}(s)$ *is a matrix that depends on the generalized curvatures* $\kappa_1(s), \ldots, \kappa_{p-1}(s)$*. The term* $v^2\boldsymbol{K}(s)$ *represents the geometric coupling between the curve's shape and the transverse stability.*

This result has a profound implication: *the stability of trajectory tracking depends on the geometry of the trajectory.* A trajectory with low curvature is inherently easier to stabilize than one with high curvature, even if the underlying system dynamics are identical. The curvature modifies the effective dynamics transverse to the path.

2.10 Fundamental Theorem and Uniqueness

We close this chapter with a theorem that is central to the trajectory-first philosophy.

Theorem 2.3 (Fundamental Theorem of Curves)**.** *Let $\kappa_1(s), \ldots, \kappa_{p-1}(s)$ be smooth functions on $[0, L]$ with $\kappa_k(s) > 0$ for $k = 1, \ldots, p-2$. Then there exists a curve $\boldsymbol{\gamma} : [0, L] \to \mathbb{R}^n$, parameterized by arc length, whose generalized curvatures are exactly $\kappa_1, \ldots, \kappa_{p-1}$. Moreover, this curve is unique up to rigid motions (translations and rotations) of $\mathbb{R}^n$.*

This theorem says: **the curvature signature is the trajectory.** Two trajectories with the same curvature functions are identical in shape. This has a practical corollary for control:

Corollary 2.4. *To specify a reference trajectory, it suffices to specify the generalized curvatures as functions of arc length, together with an initial position and orientation. This representation is:*

(i) Compact*: $p - 1$ scalar functions instead of n coordinate functions.*

(ii) Intrinsic*: independent of coordinate choice.*

(iii) Timing-independent*: the trajectory shape is fully determined without specifying when each point is reached.*

Example 2.6. Curvature Signature of a Golf Swing

A simplified 2-DOF model of the golf swing (arm + club) has a 4-dimensional state space $(\theta_{\text{arm}}, \theta_{\text{club}}, \dot{\theta}_{\text{arm}}, \dot{\theta}_{\text{club}})$. The trajectory through this space has up to 3 generalized curvatures: $\kappa_1(s)$, $\kappa_2(s)$, and $\kappa_3(s)$.
The curvature signature reveals the structure of the swing:

- $\kappa_1(s)$ peaks at the transition (the "pause at the top").
- $\kappa_2(s)$ peaks during the release, where the swing plane rotates.
- $\kappa_3(s)$ is small for a well-executed swing (the motion is approximately confined to a 3D subspace of $\mathbb{R}^4$).

A coach's instruction to "flatten your swing plane" is, in geometric terms, a request to reduce $\kappa_2(s)$ during a specific phase of the motion—a precise modification to the curvature signature.

2.11 Summary

This chapter has established the geometric language we will use throughout the book:

Concept	Control significance
Arc length s	The natural "distance along the path"—separates shape from timing
Curvature $\kappa(s)$	Determines the centripetal control demand $a_\perp = \kappa v^2$
Torsion $\tau(s)$	Measures out-of-plane complexity; relates to multi-joint coordination
Frenet–Serret frame	Provides the natural moving coordinate system for decomposing deviations into tangential and transverse components
Curvature signature	Uniquely specifies the trajectory shape, independent of timing and coordinates
Path–timing separation	Decomposes control into geometric path design and temporal speed planning

The key takeaway is that a trajectory is not just "a function of time." It is a geometric object with intrinsic structure, and this structure directly determines the difficulty and requirements of controlling it. In Chapter 3, we will see that the state spaces of mechanical systems are not flat $\mathbb{R}^n$—they are curved manifolds with their own geometry. The interplay between the geometry of the *curve* and the geometry of the *space* will produce the richest and most practically important results of this book.

2.12 Exercises

2.1. **(Computation.)** Compute the curvature and torsion of the curve $\boldsymbol{\gamma}(t) = (t, t^2, t^3)$ at $t = 0$ and $t = 1$. At which point is the curve "harder to track" at constant speed?

2.2. **(Arc length.)** The logarithmic spiral $\boldsymbol{\gamma}(t) = e^{at}(\cos t,\ \sin t)$ with $a > 0$ is a model for certain biological growth patterns.

(a) Find the arc-length function $s(t)$.

(b) Show that the curvature is $\kappa = 1/(e^{at}\sqrt{1 + a^2})$ and interpret what this means about tracking difficulty as the spiral grows.

(c) What timing law $s(t)$ would maintain constant centripetal acceleration?

2.3. **(Frenet–Serret.)** For the curve $\boldsymbol{\gamma}(s) = (\cos(s/\sqrt{2}),\ \sin(s/\sqrt{2}),\ s/\sqrt{2})$ (a helix parameterized by arc length):

(a) Verify that $\|\boldsymbol{\gamma}'(s)\| = 1$.

(b) Compute the Frenet–Serret frame $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$.

(c) Compute κ and τ and verify the Frenet–Serret equations.

2.4. (Path–timing separation.) A robot end-effector must trace the path $\boldsymbol{p}(s) = (s,\ \sin(\pi s),\ 0)$ for $s \in [0, 2]$ (one full sine wave).

(a) Compute $\kappa(s)$ and identify the points of maximum curvature.

(b) If the maximum centripetal acceleration is $a_{\max} = 10\ \mathrm{m/s}^2$, find the speed limit $v_{\max}(s)$ along the path.

(c) Design a timing law $s(t)$ that traverses the path in minimum time while respecting the speed limit. Sketch $v(s)$ and $s(t)$.

2.5. (Conceptual.) Consider two trajectories in $\mathbb{R}^4$ with identical curvature signatures $(\kappa_1(s), \kappa_2(s), \kappa_3(s))$ but different initial conditions.

(a) By the fundamental theorem, how are these trajectories related?

(b) If you designed a tracking controller for one, would it work for the other? Under what conditions?

(c) Relate this to the idea that a well-learned golf swing can be started from slightly different address positions.

2.6. (Numerical project.) Using motion capture data (or simulated data from a double pendulum), compute the curvature signature of a swing-like motion:

(a) Numerically estimate $s(t)$ by integrating $\|\dot{\boldsymbol{x}}(t)\|$.

(b) Compute $\kappa_1(s)$ using Eq. (2.13).

(c) Identify the phases of the motion (backswing, transition, downswing, impact) from features of the curvature profile.

(d) Plot the curvature cost $\kappa(s)\, v(s)^2$ and identify the most demanding phase of the motion.

The shape of a river is written in its curvature.
The story of a motion is written the same way.

Learn to read the curvature, and you read the motion.